

Designing a Research Project to Explore the Emergence of Spacetime, Particles, and Interactions from States and Relations — A Framework of Preliminary Design and Feasibility Verification for Foundational-Physics Model Exploration under Uncertainty

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Abstract

The Standard Model and General Relativity are extremely successful within their respective domains of application. At the same time, the problem of describing quantum theory and gravity naturally from a single underlying structure remains unsolved: despite decades of work on several major research programs, including superstring theory, loop quantum gravity, asymptotic safety, and holography, no completed picture uniquely recovers real low-energy physics, a quantum description of spacetime, the cosmological background, and observables at the same time [4–9]. Moreover, while the Standard Model possesses high gauge symmetry and conservation structure, the description of concrete physical states requires multiple experimentally determined parameters [10].

The author has been exploring whether spacetime, fields, particles, mass, charge, distance, and time — rather than being given from the start — can be generated spontaneously from a smaller number of “states” and “relations” alone. However, when individual candidate equations and numerical experiments are put first, the research direction tends to shift whenever a locally interesting phenomenon appears, and the exploration proceeds while the project’s goal image — “what structure must be generated for the research objective to count as achieved” — remains vague.

This paper presents no completed theory. Instead, applying project-management research on exploratory projects under high uncertainty [1–3], together with the author’s own working heuristics from IT consulting and systems design, it redesigns this research as a long-term exploratory project. Concretely, it defines an iterative research structure of philosophical principles, a provisional goal image, preliminary design, feasibility verification, numerical experiments, and design revision.

The goal of this research is not to invent a new physical law and replace existing theories. It is to explore whether there exists an underlying structure or interpretation that can explain the Standard Model and General Relativity — with their successes preserved — from fewer assumptions and fewer arbitrary parameters.

Keywords: foundational physics, quantum gravity, emergent spacetime, states and relations, exploratory project, project management, preliminary design, feasibility verification

1. Motivation

The starting point of this research is the following two questions.

First: given that the Standard Model and General Relativity are each successful to very high precision, why is it difficult to derive both naturally from the same underlying structure?

General Relativity treats spacetime geometry itself as a dynamical degree of freedom. The Standard Model, on the other hand, describes the strong, weak, and electromagnetic interactions on the basis of quantum field theory and gauge symmetry. At low energies, General Relativity can be treated quantum mechanically as an effective field theory, but no completed form of a fundamental quantum theory of gravity valid up to the Planck regime has been obtained [4,5].

Second: while the Standard Model possesses high symmetry and many conservation structures, its concrete description requires multiple experimental parameters. This does not mean the Standard Model has failed. Rather, this research considers the possibility that the existing theories are highly accurate effective descriptions, behind which there exists a foundational layer that generates the same structures from a smaller number of principles.

String theory has important achievements as a quantum framework that includes gravity, and gauge-gravity duality and holography give strong hints about the relation between spacetime and information. However, unresolved problems remain concerning background independence, non-perturbative formulation, the selection of realistic vacua, and application to cosmological spacetimes [6,7]. Loop quantum gravity, asymptotic safety, discrete spacetime models, and others have likewise shown important progress from different angles, but no consensus has been reached on a single completed theory [4,8,9].

This research does not deny these existing approaches. Rather, it treats the successes of the existing theories as strong constraints that must ultimately be reproduced.

2. Position and Scope as an Independent Researcher

The author is an independent researcher, not affiliated with a university, research institute, or academic society specializing in quantum gravity, particle theory, or general relativity.

This position has two sides.

On one hand, the author is not directly bound by the research programs, particular theoretical lineages, or community-internal problem settings usually required inside established research fields, and can explore by combining concepts from different fields and many hypotheses relatively freely. This makes it possible to test the strong hypothesis of removing the existing concepts of spacetime, fields, particles, mass, and charge themselves from the initial premises and reconstructing them from a smaller number of states and relations.

On the other hand, research resources are limited. Carrying out large-scale collaboration, dedicated computational resources, long-term rigorous mathematical analysis, and complete connection proofs to existing theories simultaneously is not realistic. For this reason, the outputs of this research primarily target the level of theoretical proposals, thought experiments, numerical experiments, and reproducible preliminary designs, and this project alone does not aim to derive a complete rigorous connection to the Standard Model or General Relativity.

This limitation does not mean neglecting consistency with existing theories. Quite the opposite. In this research, the Standard Model and General Relativity are treated not as “objects to be replaced” but as “known successes to be reproduced.”

The purpose of this research is therefore not

to invent new physics

but

to explore structures that can explain the known Standard Model and relativity from fewer assumptions and fewer arbitrary

The structure presented in this paper is meant to fix the direction of the research project toward this purpose; it is not the claim of a completed physical theory.

3. Why the Research Project Itself Is a Design Object

In this research, the author has so far repeated an exploration cycle of building individual candidate equations, running numerical experiments, and analyzing the phenomena obtained. This method is heuristically effective, but it has one problem.

When candidate equations yield interesting phenomena — singularities, closure structures, metastable states, exponential growth, periodic structures — those phenomena easily start to look as if they themselves were the research objective.

However,

an interesting numerical phenomenon $\neq$ the foundational structure actually being sought.

Project-management research has likewise pointed out that projects whose exploration methods and final goals are vague require learning, candidate selection, and iterative design revision, not merely fixed plan execution [1–3].

Turner and Cochrane classified projects by separating the clarity of goals from the clarity of the methods of achieving them [1]. Pich, Loch, and De Meyer argued for the need to use instructionism, learning, and selectionism according to the kinds of uncertainty, ambiguity, and complexity involved [2]. Lenfle showed that exploratory projects, in which neither the technology nor the requirements are fixed at the start, need management methods different from those of ordinary development projects [3].

In this sense, this research is close to a typical exploratory project.

The following iterative structure is therefore adopted:

philosophical principles $\rightarrow$ goal image $\rightarrow$ preliminary design $\rightarrow$ feasibility verification $\rightarrow$ numerical experiments $\rightarrow$ revision

What matters is that the goal image itself is not a fixed specification. If numerical experiments or mathematical examination show it to be impossible or unnatural, the goal image itself is revised.

4. First-Principle Candidate: Nameless Distinguishable States

This research keeps what exists initially to a minimum.

The first-principle candidate is expressed as follows:

There exist states that have no names, yet are mutually distinguishable and countable.

For convenience, the state set is written

$$\mathcal{Z}_n = \{z_{1,n}, z_{2,n}, \dots, z_{N_n,n}\}.$$

Here the subscripts are computational bookkeeping identifiers, not physical “names.” States are not endowed in advance with attributes such as electron, photon, mass, charge, spatial position, or time coordinate.

5. Non-direct Readability and Interaction = Readout

This research does not assume that a state itself can be read directly.

Likewise, it does not assume that a relation between states can itself be read directly.

Physical readout is taken to become possible only as a product with other states or relations.

Conceptually,

a state alone: unreadable

a relation alone: unreadable

product-type operations of states and relations: readable.

In this position, observation and interaction cannot be separated in principle.

To read anything out, an operation with other states and relations is required, and that operation itself produces the next state.

This research therefore takes

observation $\equiv$ readout accompanied by interaction.

6. Self-interaction Is Not Excluded

Since states have no physical names, there is initially no ground for excluding only oneself from the interaction partners.

Hence, writing the unknown two-state map as

$$f(z_i, z_j),$$

the minimal skeleton producing each state's next state is taken as the candidate

$$z_{i,n+1} = \sum_{j=1}^{N_n} f(z_{i,n}, z_{j,n}).$$

The sum includes

$$j = i$$

as well.

However, this paper does not decide the concrete

$$f(z_i, z_j).$$

This is precisely the main object to be explored in the coming preliminary design and feasibility verification.

7. The Project's First Goal Image

The following is neither an axiom nor a completed requirements specification.

It is the first goal image — what the author currently hopes will ultimately appear if structures resembling known physics can be generated from a small number of states and relations alone.

7.1 Global conserved quantity

The system as a whole is expected to have a conserved quantity

$$C_{\text{tot}}$$

that may later be read out as corresponding to mass or energy.

For example, a structure like

$$\boxed{\sum_{i=1}^{N_n} C(z_{i,n}) = C_{\text{tot}}.}$$

The concrete form of the conserved quantity is undetermined.

7.2 Growth and decrease of the number of states

The number of states

$$N_n$$

is not assumed fixed.

Both

$$N_{n+1} > N_n$$

and

$$N_{n+1} < N_n$$

are allowed.

A structure in which the global conserved quantity is maintained even as the number of states changes is desirable.

7.3 Extension arising from relations

Between states, multiple relational quantities

$$R_{ij}^{(\mu)}$$

are expected to form, some of which may later be read out as distance or spatial extension.

The direction μ is not given as a spatial axis from the start.

Multiple directions are expected to form spontaneously from interactions.

7.4 Initial highly symmetric state

Initially, a highly symmetric state is envisaged in which the all-direction total of the relational quantities that look like distance or extension is arbitrarily close to zero:

$$L_\mu(0) \simeq 0.$$

However, exact zero is not fixed from outside.

7.5 Spontaneous symmetry breaking and exponential extension

If the highly symmetric state is not a stable fixed point but an unstable one, a small asymmetry

$$\epsilon$$

may be amplified as

$$\epsilon \rightarrow e^{\lambda n} \epsilon,$$

producing extension without any externally specified direction.

For directions that look spatial, the appearance, over some period, of exponential growth like

$$L_s(n) \sim e^{Hn}$$

is taken as one element of the goal image.

7.6 Temporal direction

By contrast, one relational direction is expected to differ from the spatial ones: not condensing locally, but increasing monotonically as the accumulation of causal processing.

For example, an approximately linear increase like

$$L_t(n) \sim n$$

may later be read out as time.

Rather than giving time as a continuous coordinate from the start, it is desirable to generate it as the order, or accumulated amount, of discrete causal updates.

7.7 Upper bound of extension and phase transition

Spatial exponential growth is not expected to continue without limit; it is expected to reach a ceiling originating from the number of states, the number of relations, the conserved quantity, or the total of relational quantities, and to undergo another phase transition.

As a result,

$$\text{growth of wide-area relations} \rightarrow \text{condensation of local relations}$$

is considered as a possible outcome.

7.8 Condensates and particles

If locally stable relational structures form, they are read out afterwards as particles or condensates.

Particles are not a basic input.

Condensation is expected to occur in the spatial directions, while the same kind of condensation is expected not to occur in the temporal direction.

7.9 Reactions between condensates

In interactions between condensates, the conserved quantity is expected to appear not as unlimited fresh creation, but as

$$\boxed{\text{transmission} + \text{partition} + \text{redistribution.}}$$

Furthermore, it is a long-term goal that both gravitational and electromagnetic interactions later appear as different readouts of the same basic states and the same relational operations.

8. Performance Targets: Cutoff Criteria for Evaluating Candidate Models

The following are also not axioms at this stage; they are performance targets for evaluating preliminary designs.

8.1 Boundedness

The most important performance target is boundedness.

From finite input,

- state values
- relational quantities
- the number of states
- the number of relations
- the operational steps needed for one interaction

are expected not to diverge without limit.

Conceptually,

$$\boxed{\text{finite input} \rightarrow \text{finite operations} \rightarrow \text{finite output.}}$$

This differs from assuming that the universe as a whole has a permanently fixed maximum number of states. The requirement is that at any finite processing stage, the computational objects and physical quantities are finite.

8.2 Discrete causality

An interaction has the definite order

$$\text{input} \rightarrow \text{operation} \rightarrow \text{output.}$$

The basic description is therefore expected to be

$$\boxed{Z_n \rightarrow Z_{n+1}.$$

In this position, the elementary process is essentially discrete, and there is no need to postulate intermediate states

$$n + \epsilon$$

as physical reality.

Differentiable continuous equations are considered to appear, if at all, only as effective descriptions in a coarse-grained limit of many discrete updates.

8.3 Resolution floor

This research does not assume that the infinitesimal and zero are always physically distinguishable.

Readout is performed not by direct measurement of a single state, but by product-sum operations with other states and relations.

Conceptually, a finite-range product-sum readout like

$$Y_{AB} = \sum_k P(R_{Ak}, R_{Bk})$$

is considered.

If this operation acts as a finite-bandwidth integrating operation, the readout itself may possess an effective lower bound of resolution.

If the difference of two states is below the readout resolution, it is treated as

$$\boxed{\text{an unresolvable nonzero} \equiv \text{physically zero.}}$$

What matters is that this resolution is not given from the start as a fixed constant. It is expected that the two interacting relational structures themselves determine the resolution, as in

$$\boxed{\epsilon_{AB} = \mathcal{E}(R_A, R_B).}$$

Note that a low-pass characteristic does not follow automatically from “being a product-sum” alone. Whether high-frequency components are actually suppressed to yield a finite bandwidth is an item to be verified for each concrete candidate map.

9. The Position of Quantum Behavior

This research does not place quantum behavior as a first axiom.

If, from

$$\boxed{\text{discrete causal processing} + \text{finite resolution} + \text{product-sum relational readout,}}$$

distinguishable observational outcomes become naturally discretized, then quantum behavior may be explainable as an emergent property.

Accordingly, quantum behavior is treated not as

$$\boxed{\text{quantum behavior} = \text{an axiom}}$$

but as

$$\boxed{\text{quantum behavior} = \text{a phenomenon to be derived from discrete causality and finite resolution.}}$$

This is not a conclusion proven in this paper; it is a subject of future verification.

10. Connection Policy toward the Standard Model and General Relativity

This research does not start from denying or replacing the Standard Model and General Relativity.

Rather, if a candidate model matures sufficiently, it must recover at least the following two kinds of limits.

First, for gravity: reproducing the Newtonian distance dependence in the weak-field, low-velocity limit, and more generally yielding readouts consistent with the geometric description of General Relativity.

Second, for non-gravitational interactions: conservation laws, symmetries, and interaction structures corresponding to the gauge interactions of the Standard Model being read out as effective descriptions.

At the current stage of the project, however, rigorous derivation of these is not an immediate goal.

First,

does there exist a finite dynamics that can generate these structures from states and relations alone?

is examined at the level of feasibility.

In existing quantum-gravity research as well, the emergent viewpoint — classical spacetime and gravitons appearing in suitable limits from more basic short-distance structure — is an important research theme [8]. The holographic principle suggests a deep relation between geometry and information [7]. This research does not adopt these specific theories, but the exploratory direction itself — not taking known spacetime and particles as basic entities from the start — is not isolated from the problem-consciousness of contemporary quantum-gravity research.

11. Preliminary Design and Feasibility Verification

At the current stage, preliminary design and feasibility verification are needed before proceeding to basic design.

The reason is simple.

There is no guarantee that a map satisfying all the goal images listed in this paper exists.

Moreover, even if one exists, it may not be discoverable within realistic research resources.

Therefore, instead of fixing one map as the correct answer, multiple candidates

$$f_1, f_2, f_3, \dots$$

are constructed and evaluated in stages.

The first feasibility verification is limited to a small number of conditions, for example:

1. Can a global conserved quantity be maintained?
2. Can all-state interaction including self-interaction be defined?
3. Can the conservation structure be maintained as the number of states grows or shrinks?
4. Do states, relations, and operations remain finite?
5. Are relational directions generated without externally supplied spatial axes or particle species?

Candidates that fail these are not extended to the later stages of cosmic expansion, condensation, gravity, and electromagnetism.

When a candidate fails, instead of immediately adding auxiliary assumptions to keep it alive, the question

is the candidate map wrong?

or

is the goal image itself wrong?

is examined separately.

This iteration is the central method of this research.

12. Research Prohibitions

While preserving the freedom of exploration, this project avoids, as far as possible, hidden inputs such as the following.

- Giving particle-species names to initial states.
- Separating basic waves for gravity and for electromagnetism from the start.
- Fixing spatial or temporal axes externally.
- Pre-specifying the network of interaction partners without reason.
- Excluding only self-interaction without reason.
- Adding normalization or clipping solely to obtain desirable results.
- Repairing a diverging candidate with a saturation function that has no physical grounding.
- Interpreting an observed numerical phenomenon as a physical discovery without verification of correspondence with known physics.

These are not absolute prohibitions. If their necessity is derived, they may be introduced. In that case, however, they must be stated explicitly as new assumptions.

13. Success and Failure Conditions

Success for this project does not mean only obtaining a completed theory of everything.

For example, any of the following counts as an outcome:

- Showing that a minimal map exists that simultaneously satisfies part of the goal image.
- Showing that certain requirements are mathematically incompatible with each other.
- Showing that a specific known physical structure can emerge from states and relations alone.
- Conversely, showing that additional basic degrees of freedom are indispensable.
- Establishing effective exploration procedures and rejection criteria.

Therefore,

the discovery of infeasibility is also a research outcome.

14. Significance of This Paper

This paper contains no proof of a new physical law.

Its purpose is to clearly separate what has tended to be mixed together:

philosophical principles

axiom candidates

the project's goal image

assumptions of preliminary design

phenomena obtained in numerical experiments

conditions for comparison with known physics.

In foundational-physics exploration, obtaining a locally interesting phenomenon and that structure explaining real physics are not the same thing.

This project therefore emphasizes

preserving not only what was discovered, but why that computation was performed.

This is also a research-management principle for maintaining reproducibility and direction in independent research that must sustain long-term exploration with limited resources.

15. Conclusion

The immediate purpose of this research is not to present a completed equation of the universe.

More restrictedly, it is to examine

whether there exists a finite computational system that can spontaneously generate, from nameless distinguishable states,

This paper adopted no specific two-body interaction map.

Instead, it defined the research process itself:

philosophical principles → provisional goal image → preliminary design → feasibility verification → numerical experimentation

As an independent researcher, the author does not deny existing theories, but explores whether structures exist that can explain their successes from fewer assumptions.

In that process, it is accepted that candidate maps will be rejected, that the goal image itself will be revised, and that the original conception may ultimately turn out to be infeasible.

This paper is the planning document of that exploration — written so that the search neither drifts nor is fixed prematurely into a single theory.

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Note: Citation Policy of This Paper

Because the purpose of this paper is to fix the direction of the research project, the author’s own previously published papers are not used as grounds. There is no self-citation, and external citations are limited to (i) management methods for exploratory projects under high uncertainty, and (ii) reviews organizing the current position of the Standard Model, General Relativity, and quantum-gravity research.